

Math 237A HW 5

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Problem 1

Hartshorne 3.14:

Projection from a Point. Let $\mathbb{P}^n$ be a hyperplane in $\mathbb{P}^{n+1}$ and let $P \in \mathbb{P}^{n+1} \setminus \mathbb{P}^n$. Define a mapping $\varphi : \mathbb{P}^{n+1} \setminus \{P\} \rightarrow \mathbb{P}^n$ by $\varphi(Q) =$ the intersection of the unique line containing P and Q with $\mathbb{P}^n$.

- (a) Show that φ is a morphism.
- (b) Let $Y \subseteq \mathbb{P}^3$ be the twisted cubic curve which is the image of the 3-tuple embedding of $\mathbb{P}^1$ (EX. 2.12). If t, u are the homogeneous coordinates on $\mathbb{P}^1$, we say that Y is the curve given *parametrically* by $[x, y, z, w] = [t^3, t^2u, tu^2, u^3]$. Let $P = [0, 0, 1, 0]$, and let $\mathbb{P}^2$ be the hyperplane $z = 0$. Show that the projection of Y from P is a cuspidal cubic curve in the plane, and find its equation.

Solution:

- (a) First, for any $Q \in \mathbb{P}^{n+1} \setminus \mathbb{P}^n$, define the line $r_Q : \mathbb{P}^1 \rightarrow \mathbb{P}^{n+1}$ by $r_Q([s, t]) = sP + tQ = [sp_0 + tq_0, sp_1 + tq_1, \dots, sp_{n+1} + tq_{n+1}]$ as the line embedding (Note: this is well-defined because $Q \neq P$, so their representative in $\mathbb{A}^{n+2}$ are linearly independent, hence the only equation satisfying $sP + tQ = 0$ is $s = t = 0$, which is not allowed in $\mathbb{P}^1$). WLOG, define $\mathbb{P}^n := \{[x_0, x_1, \dots, x_{n+1}] \in \mathbb{P}^{n+1} | x_0 = 0\}$, then the intersection of $\mathbb{P}^n$ with the line r_Q happens when $sp_0 + tq_0 = 0$. Which, there are two cases:
 - If $Q \in \mathbb{P}^n$ (i.e. $q_0 = 0$), with $P \notin \mathbb{P}^n$, we have $p_0 \neq 0$. Then, $sp_0 + tq_0 = sp_0 = 0$ implies $s = 0$. So, the corresponding point must have $t \neq 0$, showing that $[s, t] = [0, 1] \in \mathbb{P}^1$ is the only point satisfying $r_Q([s, t]) \in \mathbb{P}^n$.
 - If $Q \notin \mathbb{P}^n$ (i.e. $q_0 \neq 0$), then notice that the linear functional $k^2 \rightarrow k$ by $(s, t) \mapsto sp_0 + tq_0$ is non-degenerate (ex: $(s, t) = (0, 1)$ has output $q_0 \neq 0$), hence this linear map has kernel being dimension 1 (i.e. kernel is $\text{span}\{u\}$ for some nonzero $u \in k^2$). So, given any $[s, t] \in \mathbb{P}^1$ satisfying $sp_0 + tq_0 = 0$, we must have $(s, t) \in \text{span}\{u\}$, showing that $[s, t] = [u]$ when taking the quotient of $\mathbb{P}^1$ as rays of k^2 passing through 0. So, $[u] \in \mathbb{P}^1$ is the unique point where $r_Q([u]) \in \mathbb{P}^n$.

With the two cases established, each line passing through P and Q must have a unique intersection, so the map $\varphi(Q) =$ the intersection of the line r_Q with $\mathbb{P}^n$ is well-defined.

To show it's in fact a morphism, notice that the solution to the above intersection (of the line through P, Q and $\mathbb{P}^n$) is given by the solution to $sp_0 + tq_0 = 0$. With $p_0 \neq 0$ (using the assumption that $P \notin \mathbb{P}^n = \{[x] \in \mathbb{P}^{n+1} | x_0 = 0\}$), we naturally have $s = -\frac{tq_0}{p_0}$. Which, we must have $t \neq 0$ (since if $t = 0$, $s = 0$, which $[s, t]$ is not well-defined in $\mathbb{P}^1$). Then, WLOG can treat $t = 1$. Hence, we have $[s, t] = \left[-\frac{q_0}{p_0}, 1\right]$.

So, the map $\varphi : \mathbb{P}^{n+1} \rightarrow \mathbb{P}^n$ is explicitly given as follow:

$$\begin{aligned}\varphi(Q) &= \frac{q_0}{p_0}P + Q = \left[-\frac{p_0}{p_0}q_0 + q_0, -\frac{p_1}{p_0}q_0 + q_1, \dots, -\frac{p_{n+1}}{p_0}q_0 + q_n \right] \\ &= \left[0, -\frac{p_1}{p_0}q_0 + q_1, \dots, -\frac{p_{n+1}}{p_0}q_0 + q_{n+1} \right] \in \mathbb{P}^n\end{aligned}\tag{1.1}$$

Since in terms of the coordinate $[q_0, \dots, q_{n+1}]$ of $\mathbb{P}^{n+1}$ the above map has all the polynomials in each entry being the same degree, it is a rational map; furthermore, since it's well-defined on $\mathbb{P}^{n+1} \setminus \{P\}$, it's in fact a well-defined morphism.

- (b) Given $P = [0, 0, 1, 0]$ and the twisted cubic curve $[x, y, z, w] = [t^3, t^2u, tu^2, u^3]$ for all $[t, u] \in \mathbb{P}^1$, and $\mathbb{P}^2 := \{[x, y, z, w] \in \mathbb{P}^3 | z = 0\}$. Then, for each $[t, u] \in \mathbb{P}^1$, the corresponding line through $[t^3, t^2u, tu^2, u^3]$ and $P = [0, 0, 1, 0]$ intersects $\mathbb{P}^2$ when $[a, b] \in \mathbb{P}^1$ satisfies $a \cdot 1 + b \cdot tu^2 = 0$, showing that the point $[a, b] = [-tu^2, 1]$ based on the logic from part (a) (where $b \neq 0$, so $b = 1$ can be chosen).

Which, the intersection of the line through P, Q and $\mathbb{P}^2$ is given by $aP + bQ$, and it's given as follow:

$$\begin{aligned}aP + bQ &= [-tu^2 \cdot 0 + t^3, -tu^2 \cdot 0 + t^2u, -tu^2 \cdot 1 + tu^2, -tu^2 \cdot 0 + u^3] \\ &= [t^3, t^2u, 0, u^3]\end{aligned}\tag{1.2}$$

Now, if consider the curve $[t^3, t^2u, u^3] \in \mathbb{P}^2$ (which is an analogy to the previous curve), using $[x, y, z]$ as coordinates of $\mathbb{P}^2$, every point on the curve satisfies the equation $x^2z = y^3$ (since we have $(t^3)^2u^3 = t^6u^3 = (t^2u)^3$). However, we can claim something even stronger, that every $[x, y, z]$ satisfying $x^2z = y^3$ is contained in the curve $\{[t^3, t^2u, u^3]\}$:

If $z = 0$, then since $y^3 = x^2z = 0$, hence $y = 0$. So, for the point $[x, y, z]$ to be well-defined, $x \neq 0$, hence can choose $x = 1$. So, the point is $[x, y, z] = [1, 0, 0]$, which belongs to the curve by choosint $[t, u] = [1, 0]$ (so $[t^3, t^2u, u^3] = [1, 0, 0]$).

Else, if $z \neq 0$, there are two cases:

- If $x = 0$, then one has $y^3 = x^2z = 0$, hence $y = 0$, showing that $[x, y, z] = [0, 0, z] = [0, 0, 1]$. So, choose $[t, u] = [0, 1]$ provides the corresponding point (since $[t^3, t^2u, u^3] = [0, 0, 1]$).
- Else if $x \neq 0$, the $y^3 = x^2z \neq 0$, showing that $y \neq 0$. So, by scaling the point to get $[x, y, z] = [\frac{x}{z}, \frac{y}{z}, 1]$, choose $[t, u] = [\frac{x}{y}, 1]$, it satisfies the following:

$$[t^3, t^2u, u^3] = \left[\frac{x^3}{y^3}, \frac{x^2}{y^2}, 1 \right] = [x^3, x^2y, y^3] = \left[x, y, \frac{y^3}{x^2} \right] = [x, y, z]\tag{1.3}$$

(Note: recall that $x^2z = y^3$, so with $x, y, z \neq 0$, $z = \frac{y^3}{x^2}$).

Hence, we concluded that $[x, y, z] \in \{[t^3, t^2u, u^3] \in \mathbb{P}^2 | [t, u] \in \mathbb{P}^1\}$ iff $x^2z = y^3$, this is the equation associated to the curve. And, notice that is the homogenization of the equation $x^2 = y^3$ in $\mathbb{A}^2$, which generates the cuspidal cubic curve in $\mathbb{A}^2$, so the curve $\{[t^3, t^2u, u^3]\}$ in fact is the closure of the cuspidal cubic curve in $\mathbb{P}^2$.

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Problem 2

Hartshorne 4.3:

- (a) Let f be the rational function on $\mathbb{P}^2$ given by $f = \frac{x_1}{x_0}$. Find the set of points where f is defined and describe the corresponding regular function.
- (b) Now think of this function as a rational map from $\mathbb{P}^2$ to $\mathbb{A}^1$. Embed $\mathbb{A}^1$ in $\mathbb{P}^1$, and let $\varphi : \mathbb{P}^2 \rightarrow \mathbb{P}^1$ be the resulting rational map. Find the set of points where φ is defined, and describe the corresponding morphism.

Solution:

- (a) In case for the rational function $f : \mathbb{P}^2 \dashrightarrow k$ by $f([x_0, x_1, x_2]) = \frac{x_1}{x_0}$ to be defined, one needs $x_0 \neq 0$. Hence, let $H_0 := \{[x_0, x_1, x_2] \in \mathbb{P}^2 \mid x_0 = 0\}$, f is defined on the open set $\mathbb{P}^2 \setminus H_0 \cong \mathbb{A}^2$ (where the isomorphism is given by the map $\mathbb{A}^1 \rightarrow \mathbb{P}^2 \setminus H_0$ by $(x, y) \mapsto [1, x, y]$).
So, if view such domain $\mathbb{P}^2 \setminus H_0$ as $\mathbb{A}^2$, the corresponding regular function is pulls back to a projection $\mathbb{A}^2 \rightarrow k$ by $(x, y) \mapsto x$, since precomposing with the isomorphism, we get the map $\mathbb{A}^2 \rightarrow \mathbb{P}^2 \setminus H_0 \rightarrow k$ by $(x, y) \mapsto [1, x, y] \mapsto \frac{x}{1} = x$.

- (b) Define the embedding $k = \mathbb{A}^1 \rightarrow \mathbb{P}^1$ by $x \mapsto [1, x]$ (hence corresponds to the open set $\mathbb{P}^1 \setminus H_0 \subseteq \mathbb{P}^1$, where $H_0 = \{[x_0, x_1] \mid x_0 = 0\}$). Compose this embedding with the previous rational map f , we get $\mathbb{P}^2 \dashrightarrow \mathbb{P}^1$ given by $[x_0, x_1, x_2] \mapsto \frac{x_1}{x_0} \mapsto \left[1, \frac{x_1}{x_0}\right]$.

Notice that on the open subset $\mathbb{P}^2 \setminus H_0$ (where $x_0 \neq 0$), such rational map is well-defined morphism, and is given by $\mathbb{P}^2 \setminus H_0 \rightarrow \mathbb{P}^1$ as $[x_0, x_1, x_2] \mapsto \left[1, \frac{x_1}{x_0}\right] = [x_0, x_1]$.

Hence, if consider the rational map $\varphi : \mathbb{P}^2 \dashrightarrow \mathbb{P}^1$ given by $[x_0, x_1, x_2] \mapsto [x_0, x_1]$, it agrees with the previous rational map $f : \mathbb{P}^2 \dashrightarrow \mathbb{A}^1 \subset \mathbb{P}^1$, and its domain in fact enlarges to the open subset where $x_0 \neq 0$ or $x_1 \neq 0$ (where $[x_0, x_1] \in \mathbb{P}^1$ is well-defined). Hence, the open subset $\mathbb{P}^2 \setminus (H_0 \cap H_1)$ (where the closed set $H_i := \{[x_0, x_1, x_2] \in \mathbb{P}^2 \mid x_i = 0\}$ for index $i = 0, 1, 2$), is the largest domain where φ is well-defined morphism, and is a projection of open subsets of $\mathbb{P}^2$ onto $\mathbb{P}^1$.

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Problem 3

Hartshorne 4.4:

A variety Y is *rational* if it is birationally equivalent to $\mathbb{P}^n$ for some n (or, equivalently by (4.5), if $k(Y)$ is a pure transcendental extension of k).

- (a) Any conic in $\mathbb{P}^2$ is a rational curve.
- (b) The cuspidal cubic $y^2 = x^3$ is a rational curve.
- (c) Let Y be the nodal cubic curve $y^2z = x^2(x+z)$ in $\mathbb{P}^2$. Show that the projection φ from the point $P = (0, 0, 1)$ to the line $z = 0$ (Ex. 3.14) induces a birational map from Y to $\mathbb{P}^1$. Thus Y is a rational curve.

Solution: First, since here we're limited to Y that's a variety, so every open subset of the given Y is guaranteed to be dense and irreducible.

- (a) Let $[x, y, z]$ be the coordinates of $\mathbb{P}^2$, for any conic in $\mathbb{P}^2$ (that's also a variety), it's the variety of the form $Y = \{ax^2 + by^2 + cz^2 + dxy + exz + fyz = 0\}$ (here can assume Y itself is not a linea embedding of $\mathbb{P}^1$). Then, fix $P = [x_0, y_0, z_0] \in Y$ and consider the projection defined in **Problem 1/Hartshorne 3.14** from P (WLOG, say $x_0 \neq 0$ so the projection is onto $\mathbb{P}^1 := \{x = 0\}$). Which, for every $[x, y, z]$ the corresponding projection is done by

$[s, t] \in \mathbb{P}^1$ that satisfies $sx_0 + tx = 0$, so $s = \frac{tx}{x_0}$, which shows that $t \neq 0$, hence up to scalar $[s, t] = \left[\frac{x}{x_0}, 1\right]$. Then, the projection point of $[x, y, z]$ onto $\mathbb{P}^1 := \{x = 0\}$ is given as:

$$sP + t[x, y, z] = \left[0, \frac{x}{x_0}y_0 + y, \frac{x}{x_0}z_0 + z\right] = [0, xy_0 + yx_0, xz_0 + zx_0] \quad (3.1)$$

Notie that such projection is injective by irreducibility of Y (since there's no more than 2 points on Y can lie on the same linear embedding of $\mathbb{P}^1$), so it suffices to find an inverse on some open subset of $\mathbb{P}^1$.

Let $[p, q]$ be the coordinates of $\mathbb{P}^1$ (where in the above the image of the projection of $[x, y, z]$ is given by $p = xy_0 + yx_0$ and $q = xz_0 + zx_0$), we can use the same line of the above projection to find an inverse. Since for some $[s, t] \in \mathbb{P}^1$ one has $sP + t[0, p, q] = [x, y, z]$ (where $[0, p, q]$ is the image of $[x, y, z]$ under projection, so the three lie on the same linear embedding of $\mathbb{P}^1$), and for the case that $[x, y, z] \neq [0, p, q]$, one can enforce that $s \neq 0$ (since $s = 0$ implies $t \neq 0$, hence $[0, p, q] = [x, y, z]$). So, on some subset one can assume $s \neq 0$, or $s = 1$. Then, the point becomes the follow:

$$sP + t[0, p, q] = [x_0, y_0 + tp, z_0 + tq] = [x, y, z] \in Y \quad (3.2)$$

So, it must satisfy the conic equation:

$$\begin{aligned} ax_0^2 + b(y_0 + tp)^2 + c(z_0 + tq)^2 + dx_0(y_0 + tp) \\ + ex_0(z_0 + tq) + f(y_0 + tp)(z_0 + tq) = 0 \end{aligned} \quad (3.3)$$

Which, since $[x_0, y_0, z_0]$ satisfies the conic equation, so the above after expansion simplifies to follow:

$$(bp^2 + cq^2 + fpq)t^2 + (2by_0p + 2cz_0q + dx_0p + ex_0q + fz_0p + fy_0q)t = 0 \quad (3.4)$$

Hence, either $t = 0$ (which corresponds to the point P , not what we want), or t being the following expression:

$$t = -\frac{2by_0p + 2cz_0q + dx_0p + ex_0q + fz_0p + fy_0q}{bp^2 + cq^2 + fpq} = \frac{F(p, q)}{G(p, q)} \quad (3.5)$$

So, since we want $t \neq 0$ (to get back to $[x, y, z]$ instead of P on the line where the projection is defined), then the only way such „reverse“ projection makes sense is on the open subset $\{F(p, q), G(p, q) \neq 0\} \subset \mathbb{P}^1$.

Finally, notice that such reverse projection $\mathbb{P}^1 \rightarrow Y$ by $[0, p, q] \mapsto [x_0, y_0 + tp, z_0 + tq]$, then with $t = \frac{F(p, q)}{G(p, q)}$ (where F, G are homogeneous with degree 1 and 2 respectively), then the expression $y_0 + tp = \frac{y_0G(p, q) + pF(p, q)}{G(p, q)}$, and $z_0 + tq = \frac{z_0G(p, q) + qF(p, q)}{G(p, q)}$ having both the numerator and denominator being both homogeneous polynomials of degree 2, hence it's a well-defined rational function.

Which, for all $[x, y, z] \in Y \setminus \{P\}$, one can check that $[x, y, z] \mapsto [0, xy_0 + yx_0, xz_0 + zx_0] = [0, p, q] \in \mathbb{P}^1$ has reverse projection outputting $[x, y, z]$ (if both $F(p, q), G(p, q) \neq 0$), and similarly any $[0, p, q]$ on the open subset $\{F(p, q), G(p, q) \neq 0\}$ has been checked that the reverse projection then the projection provides $[0, p, q]$. So, up to suitable restriction on some open subset of Y and $\mathbb{P}^1$ respectively, one can say the projection and the reverse

projection are mutual inverse, hence Y is birationally equivalent to $\mathbb{P}^1$, which is a rational curve.

(Note: The computation listing here will be too long, including in here adds an extra 5 to 6 pages, which I'll temporarily skip unless it's required...then I'll add it back).

- (b) Since the cuspidal cubic $y^2 = x^3$ can be viewed as an affine variety in $\mathbb{A}^2$, it suffices to show that it's birationally equivalent to $\mathbb{A}^1 \subseteq \mathbb{P}^1$. Here, let t be the coordinate of $\mathbb{A}^1$, and (x, y) be the coordinates of $\mathbb{A}^2$.

Given any $t \in \mathbb{A}^1$, the morphism $\varphi : \mathbb{A}^1 \rightarrow \mathbb{A}^2$ by $\varphi(t) = (t^2, t^3)$ has the image satisfies the equation $y^2 = (t^3)^2 = t^6 = (t^2)^3 = x^3$, hence belong to the cuspidal cubic.

Now, restrict the domain to open subsets $\mathbb{A}^1 \setminus \{0\} \subset \mathbb{A}^1$ and $Y \setminus \{x = 0\} \subset \mathbb{A}^2$ (where $Y = \{(x, y) \in \mathbb{A}^2 | y^2 = x^3\}$), notice the map becomes a rational isomorphism:

The map $\psi : \mathbb{A}^2 \dashrightarrow \mathbb{A}^1$ given by $\psi(x, y) = \frac{y}{x}$ is well-defined whenever $x \neq 0$. Hence, it is well-defined on $Y \setminus \{x = 0\}$ (since all points with $x = 0$ is removed). And, for any $t \in \mathbb{A}^1 \setminus \{0\}$ and $(x, y) \in Y \setminus \{x = 0\}$, the following two equations are true:

$$\psi \circ \varphi(t) = \psi(t^2, t^3) = \frac{t^3}{t^2} = t \quad (3.6)$$

$$\varphi \circ \psi(x, y) = \varphi\left(\frac{y}{x}\right) = \left(\frac{y^2}{x^2}, \frac{y^3}{x^3}\right) = \left(\frac{x^3}{x^2}, \frac{y^3}{y^2}\right) = (x, y) \quad (3.7)$$

The last equality holds because of the equation of Y given by $y^2 = x^3$. Hence, we've shown that $Y \subset \mathbb{A}^2$ the cuspidal cubic is birationally equivalent to $\mathbb{A}^1 \setminus \{0\}$.

Lastly, if consider $\mathbb{A}^1 := \mathbb{P}^1 \setminus H_0$ (where $H_0 := \{[u, t] \in \mathbb{P}^1 | u = 0\} = \{[0, 1]\}$). Since WLOG can assume $u = 1$, each $[u, t] \in \mathbb{P}^1 \setminus H_0$ can be identified just by t ; then, since $0 \in \mathbb{A}^1$ is identified by the point $[1, 0] \in \mathbb{P}^1 \setminus H_0$, the corresponding open set $\mathbb{A}^1 \setminus \{0\} = (\mathbb{P}^1 \setminus H_0) \setminus \{[1, 0]\} = \mathbb{P}^1 \setminus (H_0 \cup \{[1, 0]\})$ is an open subset of $\mathbb{P}^1$. Hence, the cuspidal cubic $Y \subset \mathbb{A}^2$ is birationally equivalent to $\mathbb{P}^1$.

- (c) Given any $[x, y, z] \in \mathbb{P}^2 \setminus \{P\}$ where $P = [0, 0, 1]$, the projection onto $\mathbb{P}^1 := \{[x, y, z] \in \mathbb{P}^2 | z = 0\}$ is given by a unique $[s, t] \in \mathbb{P}^1$ satisfying $s \cdot 1 + tz = 0$ (proved in **Problem 1, Hartshorne 3.14**). Which, it is given by $s = tz$, where t can be chosen as 1, so $[s, t] = [z, 1] \in \mathbb{P}^1$ is the point satisfying the property, which induces the projection of $[x, y, z]$ onto $sP + t[x, y, z] = [0 + x, 0 + y, z - z] = [x, y, 0] \in \mathbb{P}^1$. So, let $Y = \{y^2z = x^2(x + z)\}$ be the nodal cubic curve (**Rmk:** $P = [0, 0, 1] \in Y$, but the projection is invalid at this point), then the previous projection when excluding P , has $\pi : Y \dashrightarrow \mathbb{P}^1$ by $\pi([x, y, z]) = [x, y]$ being a well-defined rational map on $Y \setminus \{P\}$ (since the only point when it's not well-defined, is when both $x, y = 0$, yet this enforces $[x, y, z] = [0, 0, 1] = P$, hence excluding P makes π well-defined).

Now, to find a rational inverse, consider the open subset $\mathbb{P}^1 \setminus \{y^2 - x^2 = 0\}$ of $\mathbb{P}^1$, and consider the rational map $\psi : \mathbb{P}^1 \dashrightarrow Y$ by $\psi([x, y]) = [x(y^2 - x^2), y(y^2 - x^2), x^3]$. First, on open subset $\mathbb{P}^1 \setminus \{y^2 - x^2 = 0\}$ this map is well-defined, since any point $[x, y] \in \mathbb{P}^1 \setminus \{y^2 - x^2 = 0\}$ satisfies $y^2 - x^2 \neq 0$, and either $x \neq 0$ or $y \neq 0$, hence either $x(y^2 - x^2) \neq 0$ or $y(y^2 - x^2) \neq 0$, so $\psi([x, y]) \in \mathbb{P}^2$; also, since $y^2 - x^2 \neq 0$, the output satisfies the following:

$$\psi([x, y]) = [x(y^2 - x^2), y(y^2 - x^2), x^3] = \left[x, y, \frac{x^3}{y^2 - x^2} \right] \quad (3.8)$$

Which, such point satisfies:

$$y^2 \left(\frac{x^3}{y^2 - x^2} \right) = x^2 \left(\frac{x(y^2)}{y^2 - x^2} \right) = x^2 \left(\frac{x(y^2 - x^2) + x^3}{y^2 - x^2} \right) = x^2 \left(x + \frac{x^3}{y^2 - x^2} \right) \quad (3.9)$$

So, let $z = \frac{x^3}{y^2 - x^2}$, $\psi([x, y]) = [x, y, z]$ satisfies $y^2 z = x^2(x + z)$, showing that $\psi([x, y]) \in Y$. So, $\psi : \mathbb{P}^1 \dashrightarrow Y$ is indeed a well-defined rational map on open subset $\mathbb{P}^1 \setminus \{y^2 - x^2 = 0\}$.

Finally, given $\mathbb{P}^1 \setminus \{y^2 - x^2 = 0\} \subset \mathbb{P}^1$ as open subset, and $Y \setminus (\{P\} \cup \{y^2 - x^2 = 0\}) \subset Y$ as another open subset, one can show π, ψ are indeed mutual inverses: We've already shown that $\pi : Y \dashrightarrow \mathbb{P}^1$ is well-defined on $Y \setminus \{P\}$, hence also well-defined on $Y \setminus (\{P\} \cup \{y^2 - x^2 = 0\})$; also, $\psi : \mathbb{P}^1 \dashrightarrow Y$ is well-defined on $\mathbb{P}^1 \setminus \{y^2 - x^2 = 0\}$.

Then, given any $[x, y, z] \in Y \setminus (\{P\} \cup \{y^2 - x^2 = 0\})$ (satisfying $x \neq 0$ or $y \neq 0$, with extra condition $y^2 - x^2 \neq 0$), it satisfies:

$$\pi([x, y, z]) = [x, y] \in \mathbb{P}^1 \setminus \{y^2 - x^2 = 0\} \quad (3.10)$$

$$\psi \circ \pi([x, y, z]) = \psi([x, y]) = \left[x, y, \frac{x^3}{y^2 - x^2} \right] \quad (3.11)$$

Since $y^2 - x^2 \neq 0$, then with the point satisfying $y^2 z = x^2(x + z)$, we have $(y^2 - x^2)z = x^3$, or $z = \frac{x^3}{y^2 - x^2}$, showing that $\psi \circ \pi([x, y, z]) = \left[x, y, \frac{x^3}{y^2 - x^2} \right] = [x, y, z]$, which $\psi \circ \pi = \text{id}$ on $Y \setminus (\{P\} \cup \{y^2 - x^2 = 0\})$.

Similarly, any $[x, y] \in \mathbb{P}^1 \setminus \{y^2 - x^2 = 0\}$ (with $x \neq 0$ or $y \neq 0$, together with $y^2 - x^2 \neq 0$), it satisfies:

$$\psi([x, y]) = \left[x, y, \frac{x^3}{y^2 - x^2} \right] \in Y \setminus (\{P\} \cup \{y^2 - x^2 = 0\}) \quad (3.12)$$

$$\pi \circ \psi([x, y]) = \pi \left(\left[x, y, \frac{x^3}{y^2 - x^2} \right] \right) = [x, y] \quad (3.13)$$

Hence, $\pi \circ \psi = \text{id}$ on $\mathbb{P}^1 \setminus \{y^2 - x^2 = 0\}$ also.

This shows that π, ψ are indeed mutual inverse on these restricted open subsets, hence Y and $\mathbb{P}^1$ are indeed birationally equivalent, Y is a rational curve.

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Problem 4

Hartshorne 4.6:

Plane Cremona Transformations. A birational map of $\mathbb{P}^2$ into itself is called a *plane Cremona transformation*. We give an example, called a *quadratic transformation*. It is the rational map $\varphi : \mathbb{P}^2 \dashrightarrow \mathbb{P}^2$ given by $(a_0, a_1, a_2) \rightarrow (a_1a_2, a_0a_2, a_0a_1)$ when no two of a_0, a_1, a_2 are 0.

- (a) Show that φ is birational, and is its own inverse.
- (b) Find open sets $U, V \subseteq \mathbb{P}^2$, such that $\varphi : U \rightarrow V$ is an isomorphism.
- (c) Find the open sets where φ and φ^{-1} are defined, and describe the corresponding morphisms. See also (V, 4.2.3).

Solution:

- (a) Let $[x_0, x_1, x_2]$ be the coordinates of $\mathbb{P}^2$, and let $H_i := \{x_i = 0\}$ be the corresponding algebraic set of index $i = 0, 1, 2$. Then, define $U_i := \mathbb{P}^2 \setminus H_i$, consider the open set $U := U_0 \cap U_1 \cap U_2 = \mathbb{P}^2 \setminus (H_0 \cup H_1 \cup H_2)$ (where every $[x_0, x_1, x_2] \in U$ has $x_0, x_1, x_2 \neq 0$). Then, if consider φ when restricting onto U , we have $\varphi([x_0, x_1, x_2]) = [x_1x_2, x_0x_2, x_0x_1] \in U$ (since each of the component $x_i \neq 0$, then their finite product is still nonzero. Hence, each entry of the output is nonzero).

Now, notice that $[x_0, x_1, x_2] \in U$ implies $\varphi([x_0, x_1, x_2]) = [x_1x_2, x_0x_2, x_0x_1] = \left[\frac{1}{x_0}, \frac{1}{x_1}, \frac{1}{x_2}\right]$ (since $x_0x_1x_2 \neq 0$, so divide every entry by it). Then, $\varphi^2 : U \rightarrow U$ satisfies $\varphi^2([x_0, x_1, x_2]) = \varphi\left(\left[\frac{1}{x_0}, \frac{1}{x_1}, \frac{1}{x_2}\right]\right) = \left[\frac{1}{1/x_0}, \frac{1}{1/x_1}, \frac{1}{1/x_2}\right] = [x_0, x_1, x_2]$. Hence, when restrict onto the open subset U , φ is its own mutual inverse, which is a birational equivalence.

- (b) Based on the statement established in (a), choose $U = V = \mathbb{P}^2 \setminus (H_0 \cup H_1 \cup H_2)$, then $\varphi : U \xrightarrow{\sim} V$ is an isomorphism with itself being its own inverse (since it's well-defined on each point of $U = V$, while also having an inverse on this open set).
- (c) In case for φ to be well-defined, then at least one of the entries in the output needs to be nonzero. For definiteness, say $\varphi([x_0, x_1, x_2]) = [x_1x_2, x_0x_2, x_0x_1]$ has the first coordinate being nonzero, then $x_1x_2 \neq 0$ implies $x_1, x_2 \neq 0$. Apply similar logic to the other two coordinates, one can deduce that on the open set where φ is well-defined, one must have at least two coordinates being nonzero. Also, conversely if two of the coordinates are nonzero (for definiteness, say $x_1, x_2 \neq 0$), then $\varphi([x_0, x_1, x_2]) = [x_1x_2, x_0x_2, x_0x_1]$ has the first coordinate being nonzero, which is well-defined.

So, φ being well-defined at $[x_0, x_1, x_2] \in \mathbb{P}^2$ iff two of its coordinates are nonzero. Hence, $\varphi([x_0, x_1, x_2])$ is well-defined iff $[x_0, x_1, x_2] \in W = (U_0 \cap U_1) \cup (U_0 \cap U_2) \cup (U_1 \cap U_2)$, where each open set $U_i := \mathbb{P}^2 \setminus H_i = \{x_i \neq 0\}$. This will be the open set where φ is well-defined.

Besides the collection $U_0 \cap U_1 \cap U_2 = U$ (where U defined in part (a) has φ as a self-isomorphism on U), there are other sets (not necessarily open in $\mathbb{P}^2$) that is within the domain W . Given $W \setminus U$, it is the collection of points in W (with at least 2 nonzero coordinates) that're not in U (not all 3 coordinates are nonzero), which there are exactly 2 coordinates being nonzero. So, $W \setminus U = \bigcup_{i=0}^2 W \cap H_i$ (which each $W \cap H_i$ is the set with $x_i = 0$, and the other two coordinates being nonzero).

Given each $W \cap H_i$ (for definiteness, we'll demonstrate $i = 0$), since $x_0 = 0$ and $x_1, x_2 \neq 0$, then one has $\varphi([0, x_1, x_2]) = [x_1x_2, 0 \cdot x_2, 0 \cdot x_1] = [1, 0, 0] \in \mathbb{P}^2$. Apply similar logic to other index i , one can deduce that any $W \cap H_i$ has all its points being sent to the point where $x_i = 1$, and other coordinates are 0.

So as conclusion, $\varphi : W \rightarrow \mathbb{P}^2$ has $\varphi : U \xrightarrow{\sim} U$ given by $\varphi([x_0, x_1, x_2]) = \left[\frac{1}{x_0}, \frac{1}{x_1}, \frac{1}{x_2}\right]$; also, $\varphi : W \cap H_0 \rightarrow \mathbb{P}^2$ is given by $\varphi([x_0, x_1, x_2]) = [1, 0, 0]$, $\varphi : W \cap H_1 \rightarrow \mathbb{P}^2$ by $\varphi([x_0, x_1, x_2]) = [0, 1, 0]$, and $\varphi : W \cap H_2 \rightarrow \mathbb{P}^2$ by $\varphi([x_0, x_1, x_2]) = [0, 0, 1]$.